ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
SEMESTER 3

DISCRETE MATHEMATICS

Module 1 – Mathematical Logic

NOTE - Part - 1

Topics: Discrete vs Continuous Mathematics • Propositions • Logical Connectives • Truth Tables • Tautology • Contradiction • Contingency • Logical Equivalence • Converse • Inverse • Contrapositive • Duality Principle

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1. Discrete vs Continuous Mathematics

1.1 What is Discrete Mathematics?

> **Definition:**
> Discrete Mathematics is the branch of mathematics that studies **countable, distinct and separate objects** rather than continuous quantities.

Discrete Mathematics provides an important mathematical foundation for **Computer Science, Artificial Intelligence, Machine Learning, digital systems, cryptography, algorithms, databases and data structures.**

A computer works with separate and countable pieces of information such as **0 and 1, True and False, Yes and No.** Therefore, mathematics dealing with such separate values is called **Discrete Mathematics.**

1.2 Simple Example

Imagine a classroom.

You can count:

- 45 students
- 30 chairs
- 5 windows

These are **discrete** because they are separate objects that can be counted individually.

But consider:

- Air in the room
- Heat
- Time passing

These quantities are treated as **continuous** because they can vary smoothly and can take values within a range.

1.3 Discrete Mathematics and Computers

> **Important Idea:**

Every computer program makes decisions.

For example:

IF password is correct → Open account
ELSE → Show "Invalid Password"

How does the computer make such a decision?

Answer: Using Logic.

1.4 Discrete vs Continuous Mathematics

Continuous Mathematics	Discrete Mathematics
Deals with continuous values	Deals with separate and countable values
Example: Time, Distance, Temperature	Example: Students, Books, Cars
Uses concepts such as Calculus	Uses Logic, Sets, Graphs, etc.

Remember:

Continuous → quantities that vary smoothly.

Discrete → separate and countable objects or values.

2. Propositions / Statements

2.1 Definition of Proposition

Definition:
A proposition is a **declarative sentence** that is either **True or False, but not both.**

The truth value of a proposition tells us whether the proposition is **True (T)** or **False (F)**.

$$\text{Truth Value} \in \{T, F\}$$

Propositions are usually represented using lowercase letters such as **p, q, r, ...**

2.2 Examples of Propositions

Sentence	Proposition?	Reason / Truth Value
2 + 3 = 5	Yes	True
Earth is round.	Yes	True
10 > 20	Yes	False
Water boils at 100°C at standard atmospheric pressure.	Yes	True

2.3 Examples That Are NOT Propositions

Sentence	Reason
How are you?	Question

Close the door.	Command
x + 5 = 10	Truth depends on the value assigned to x
Wow!	Exclamation

Important: A sentence does not become a proposition merely because it ends with a full stop. The important condition is that it must be a **declarative sentence with a definite truth value.**

3. Simple and Compound Propositions

3.1 Simple Proposition

A **simple proposition** contains one basic statement and cannot be divided into smaller propositions using logical connectives.

Example:

p : It is raining.

3.2 Compound Proposition

A **compound proposition** is formed by combining two or more propositions using logical connectives.

Let:

p : It is raining.

q : I carry an umbrella.

Then:

$$p \wedge q$$

means:

"It is raining AND I carry an umbrella."

4. Logical Connectives

Definition:
Logical connectives are symbols used to combine or modify propositions to form compound propositions.

The five important logical connectives are:

1. Negation (NOT)
2. Conjunction (AND)
3. Disjunction (OR)
4. Implication (IF...THEN)
5. Biconditional (IF AND ONLY IF)

4.1 Negation (NOT)

Symbol: ¬p

Meaning: The opposite or negation of p.

Example:

p : It is raining.

¬p : It is not raining.

p	¬p
T	F
F	T

4.2 Conjunction (AND)

Symbol: $p \wedge q$

Meaning: Both propositions must be true.

Example:

"I studied AND I passed."

p : I studied.

q : I passed.

Therefore:

$$p \wedge q$$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Remember:
Only **TT** gives True for AND.

4.3 Disjunction (OR)

Symbol: p ∨ q

Meaning: At least one of the propositions is true.

Example:

"I will have tea OR I will have coffee."

p	q	p ∨ q
T	T	T
T	F	T
F	T	T
F	F	F

Remember:
Only **FF** gives False for OR.

4.4 Implication

Symbol: p → q

Read as: "If p, then q."

Example:

"If it rains, then roads become wet."

p : It rains.

q : Roads become wet.

Therefore:

$$p \rightarrow q$$

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

Golden Rule of Implication:

p → q is **False only when p is True and q is False.**

$$T \rightarrow F = F$$

4.5 Biconditional

Symbol: p ↔ q

Read as: "p if and only if q."

Example:

"You pass the exam if and only if you study."

p : You pass the exam.

q : You study.

Therefore:

$$p \leftrightarrow q$$

p	q	p ↔ q
T	T	T
T	F	F
F	T	F
F	F	T

Remember: Biconditional is True when both propositions have the **same truth value.**

TT → True
FF → True

4.6 Summary of Logical Connectives

Connective	Symbol	Meaning	When is it True?

Negation	¬p	NOT p	Opposite of p
Conjunction	p ∧ q	p AND q	Only TT
Disjunction	p ∨ q	p OR q	TT, TF, FT
Implication	p → q	If p, then q	All except TF
Biconditional	p ↔ q	p iff q	TT and FF

5. Truth Tables

> **Definition:**
> A truth table is a table that shows all possible combinations of truth values of the propositions involved in a logical expression.

5.1 Number of Rows

If a truth table contains **n distinct propositions**, then the number of rows required is:

$$\text{Number of rows} = 2^n$$

Number of Propositions	Number of Rows
1	$2^1 = 2$

2	$2^2 = 4$
3	$2^3 = 8$
4	$2^4 = 16$

Example:

Suppose an expression contains p, q and r.

Here, n = 3.

Therefore:

$$\text{Number of rows} = 2^3 = 8$$

6. Tautology, Contradiction and Contingency

6.1 Tautology

A **tautology** is a compound proposition that is **always True**, regardless of the truth values of its component propositions.

Example: p ∨ ¬p

p	¬p	p ∨ ¬p
T	F	T

F	T	T

The last column contains only True values.

Therefore:

$$p \vee \neg p = \text{Tautology}$$

6.2 Contradiction

A **contradiction** is a compound proposition that is **always False**, regardless of the truth values of its component propositions.

Example: p ∧ ¬p

p	¬p	p ∧ ¬p
T	F	F
F	T	F

The last column contains only False values.

Therefore:

$$p \wedge \neg p = \text{Contradiction}$$

6.3 Contingency

A **contingency** is a compound proposition that is **True for some combinations** of truth values and **False for other combinations.**

Example: $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The final column contains both True and False values.

$$p \wedge q = \text{Contingency}$$

6.4 Comparison

Type	Last Column of Truth Table
Tautology	All True
Contradiction	All False
Contingency	Mixture of True and False

7. Translating English Sentences into Logical Expressions

Basic Rule:

Before translating an English sentence into a logical expression, first identify:

1. The simple propositions.
2. The connecting word or phrase.
3. The corresponding logical symbol.

7.1 Common English Words and Logical Symbols

English Word / Phrase	Logical Symbol	Example
Not	¬	Not raining → ¬p
And	∧	Study and pass → p ∧ q
Or	∨	Tea or coffee → p ∨ q
If...then	→	If p then q → p → q
If and only if	↔	p iff q → p ↔ q

7.2 Example 1 – AND

Let:

p : I study.

q : I pass.

Sentence:

I study and I pass.

Connecting word: AND

Expression:

$$p \wedge q$$

7.3 Example 2 – OR

Sentence:

I study or I pass.

Expression:

$$p \vee q$$

7.4 Example 3 – NOT

Sentence:

I do not study.

Expression:

$$\neg p$$

7.5 Example 4 – IF...THEN

Sentence:

If I study, then I pass.

Let:

p : I study.

q : I pass.

Therefore:

$$p \rightarrow q$$

7.6 Understanding "Only If"

Rule:

$$p \text{ only if } q : p \rightarrow q$$

Example:

"I pass only if I study."

Let:

p : I pass.

q : I study.

Therefore:

$$p \rightarrow q$$

This means:

"If I pass, then I study."

PASS → STUDY

7.7 Understanding "If and Only If"

Rule:

$$p \text{ if and only if } q : p \leftrightarrow q$$

"If and only if" represents a **two-way relationship.**

It means both:

$$p \rightarrow q$$

AND

$$q \rightarrow p$$

Therefore:

$$p \leftrightarrow q$$

$$\mathbf{p \leftrightarrow q}$$

7.8 Only If vs If and Only If

Phrase	Meaning	Direction
Only if	One condition is necessary for another	One-way
If and only if	Both conditions imply each other	Two-way

Only If:

I write the exam only if I paid the fee.

$$\text{Write Exam} \rightarrow \text{Paid Fee}$$

There is only one direction.

If and Only If:

I pass if and only if I receive the required result.

$$\text{Pass} \leftrightarrow \text{Receive Required Result}$$

Both directions are involved.

7.9 Example – Compound Premise

Let:

p : I study.

q : I attend class.

r : I pass.

Sentence:

If I study and attend class, then I pass.

Step 1: Identify the first part.

$$\text{I study and attend class} = p \land q$$

Step 2: Identify "if...then".

$$\rightarrow$$

Step 3: Form the complete expression.

$$(p \land q) \rightarrow r$$

Why are parentheses used?

Parentheses show that **p ∧ q** forms one complete part of the implication.

8. Logical Equivalence

Definition:

Two propositions are logically equivalent if they have **identical truth values for every possible combination** of truth values of their variables.

Notation:

$$p \equiv q$$

Read this as:

"p is logically equivalent to q."

8.1 Example – Double Negation

Consider:

$$\neg(\neg p)$$

p	¬p	¬¬p
T	F	T
F	T	F

Compare the first column and the third column.

They are identical.

Therefore:

$$\neg(\neg p) \equiv p$$

9. Important Logical Equivalences

Law	Logical Equivalence
Double Negation	$\neg(\neg p) \equiv p$
Identity	$p \wedge T \equiv p$ $p \vee F \equiv p$
Domination	$p \vee T \equiv T$ $p \wedge F \equiv F$
Idempotent	$p \vee p \equiv p$ $p \wedge p \equiv p$
Complement	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$
De Morgan's Law	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
De Morgan's Law	$\neg(p \vee q) \equiv \neg p \wedge \neg q$

Memory Tip for De Morgan's Laws:

When a negation moves inside parentheses:

- AND changes to OR.
- OR changes to AND.

- Each proposition is negated.

10. Converse, Inverse and Contrapositive

Consider the original implication:

$$p \rightarrow q$$

Example:

If it rains, then roads are wet.

Let:

p : It rains.

q : Roads are wet.

10.1 Original Statement

$$p \rightarrow q$$

If it rains, then roads are wet.

10.2 Converse

Rule:

Interchange p and q.

$$p \rightarrow q \text{ becomes } q \rightarrow p$$

Therefore:

If roads are wet, then it rains.

The converse is **not necessarily logically equivalent** to the original implication.

10.3 Inverse

Rule:

Negate both p and q.

$$p \to q \text{ becomes } \neg p \to \neg q$$

Therefore:

If it does not rain, then roads are not wet.

The inverse is **not necessarily logically equivalent** to the original implication.

10.4 Contrapositive

Rule:

Interchange p and q and negate both.

$$p \to q \text{ becomes } \neg q \to \neg p$$

Therefore:

If roads are not wet, then it did not rain.

Important: The contrapositive is logically equivalent to the original implication.

$$p \to q \equiv \neg q \to \neg p$$

10.5 Summary

Form	Symbol	How to Form
Original	$p \to q$	Given statement

Converse	$q \rightarrow p$	Interchange p and q
Inverse	$\neg p \rightarrow \neg q$	Negate both
Contrapositive	$\neg q \rightarrow \neg p$	Interchange and negate

Memory Trick:

Original: **p → q**

Converse: **q → p**

Inverse: **¬p → ¬q**

Contrapositive: **¬q → ¬p**

11. Duality Principle

Definition:

The dual of a logical expression is obtained by replacing:

- AND (∧) with OR (∨)
- OR (∨) with AND (∧)
- True (T) with False (F)
- False (F) with True (T)

The variables such as p, q, r and the negation symbol ¬ are **not changed.**

11.1 Example 1

Find the dual of:

$$p \wedge T$$

Replace:

$$\begin{array}{c} \wedge \rightarrow \vee \\ T \rightarrow F \end{array}$$

Therefore:

$$\text{Dual} = p \vee F$$

11.2 Example 2

Find the dual of:

$$(p \wedge q) \vee r$$

Replace:

- $\wedge \rightarrow \vee$
- $\vee \rightarrow \wedge$

Therefore:

$$\text{Dual} = (p \vee q) \wedge r$$

11.3 Example 3

Find the dual of:

$$(p \vee q) \wedge (r \vee T)$$

Replace:

- First $\vee \rightarrow \wedge$
- $\wedge \rightarrow \vee$
- Second $\vee \rightarrow \wedge$
- $T \rightarrow F$

Therefore:

$$\text{Dual} = (p \wedge q) \vee (r \wedge F)$$

Important:

While finding the dual:

- Do NOT change p, q, r.
- Do NOT change negation ¬.
- Change ∧ to ∨.
- Change ∨ to ∧.
- Change T to F.
- Change F to T.

12. Common Mistakes to Avoid

1. **Thinking every sentence is a proposition.**
   Questions, commands and exclamations are generally not propositions.
2. **Confusing implication with biconditional.**
   $p \rightarrow q$ means one direction, while $p \leftrightarrow q$ represents both directions.
3. **Forgetting the special case of implication.**
   $p \rightarrow q$ is False only when p = T and q = F.
4. **Reversing "only if".**
   "p only if q" means $p \rightarrow q$.
5. **Confusing converse and contrapositive.**
   Converse: $q \rightarrow p$.
   Contrapositive: $\neg q \rightarrow \neg p$.
6. **Changing variables while finding a dual.**
   Variables remain unchanged.
7. **Using the wrong number of truth-table rows.**
   For n propositions, always use 2^n rows.

13. Quick Revision Summary

Concept	Key Point
Discrete Mathematics	Studies separate and countable objects.
Proposition	Declarative sentence that is either True or False.
Negation	$\neg p$
Conjunction	$p \wedge q$; True only when both are True.
Disjunction	$p \vee q$; False only when both are False.
Implication	$p \rightarrow q$; False only for $T \rightarrow F$.
Biconditional	$p \leftrightarrow q$; True when both have the same truth value.
Truth Table	Shows all possible truth-value combinations.
Number of rows	2^n

Tautology	Always True.
Contradiction	Always False.
Contingency	Sometimes True and sometimes False.
Logical Equivalence	Same truth values in every possible case.
Converse	$q \rightarrow p$
Inverse	$\neg p \rightarrow \neg q$
Contrapositive	$\neg q \rightarrow \neg p$
Duality	Interchange $\wedge \leftrightarrow \vee$ and $T \leftrightarrow F$.

14. Practice Questions

A. Very Short Answer Questions

1. Define Discrete Mathematics.
2. What is a proposition?
3. What is the truth value of a proposition?
4. Give two examples of propositions.
5. Give two examples of sentences that are not propositions.
6. What is a simple proposition?

7. What is a compound proposition?
8. Write the symbol for negation.
9. Write the symbol for conjunction.
10. Write the symbol for disjunction.
11. Write the symbol for implication.
12. Write the symbol for biconditional.
13. What is a tautology?
14. What is a contradiction?
15. What is a contingency?

B. Truth Table Questions

1. Construct the truth table for $p \wedge q$.
2. Construct the truth table for $p \vee q$.
3. Construct the truth table for $p \rightarrow q$.
4. Construct the truth table for $p \leftrightarrow q$.
5. Construct the truth table for $p \vee \neg p$ and identify its type.
6. Construct the truth table for $p \wedge \neg p$ and identify its type.

C. Translation Questions

1. Let p = "I study" and q = "I pass". Translate: "I study and I pass."
2. Translate: "I study or I attend class."
3. Translate: "I do not study."
4. Translate: "If I study, then I pass."
5. Translate: "I pass only if I study."
6. Translate: "I study if and only if I attend class."
7. Let p = "I study", q = "I attend class", r = "I pass". Translate: "If I study and attend class, then I pass."

D. Converse, Inverse and Contrapositive

For each statement, write its converse, inverse and contrapositive.

1. If a number is divisible by 2, then it is even.
2. If it rains, then the ground becomes wet.
3. If a student studies regularly, then the student performs well.

E. Logical Equivalence

1. Show that ¬(¬p) ≡ p using a truth table.
2. Verify p ∨ ¬p is a tautology.
3. Verify p ∧ ¬p is a contradiction.
4. Verify one of De Morgan's laws using a truth table.

F. Duality Principle

1. Find the dual of p ∧ T.
2. Find the dual of p ∨ F.
3. Find the dual of (p ∧ q) ∨ r.
4. Find the dual of (p ∨ q) ∧ (r ∨ T).

15. Exam Tips

Tip 1 – Remember the truth-table rules

- AND → only TT is True.
- OR → only FF is False.
- Implication → only TF is False.
- Biconditional → TT and FF are True.

Tip 2 – Remember the classification

- All T → Tautology
- All F → Contradiction
- Both T and F → Contingency

Tip 3 – Remember the four forms

$$\text{Original: } p \to q$$

$$\text{Converse: } q \to p$$

$$\text{Inverse: } \neg p \to \neg q$$

$$\text{Contrapositive: } \neg q \rightarrow \neg p$$

Tip 4 – Remember "Only If"

$$p \text{ only if } q : p \rightarrow q$$

The statement before **only if** becomes the antecedent.

Tip 5 – Remember "If and Only If"

$$p \text{ if and only if } q : p \leftrightarrow q$$

It represents two directions:

$$p \rightarrow q$$
$$\text{AND}$$
$$q \rightarrow p$$

16. Module 1 – Part 1: Final Takeaway

Discrete Mathematics deals with mathematical objects that are **separate, distinct and countable.**

A **proposition** is a declarative sentence that is either True or False.

Propositions can be combined using logical connectives such as **¬, ∧, ∨, → and ↔.**

Truth tables help us determine the truth value of compound propositions for every possible combination of input values.

A proposition is:

- **Tautology** if it is always True.
- **Contradiction** if it is always False.
- **Contingency** if it is sometimes True and sometimes False.

Logical equivalence means that two logical expressions have the same truth value for every possible combination of truth values.

For an implication p → q, we can form the **converse, inverse and contrapositive.** The contrapositive is logically equivalent to the original implication.

The **duality principle** allows us to obtain the dual of a logical expression by interchanging AND with OR and True with False.

Note: This document covers only **Part 1 of Module 1**. The remaining topics of Module 1 will be covered separately.
